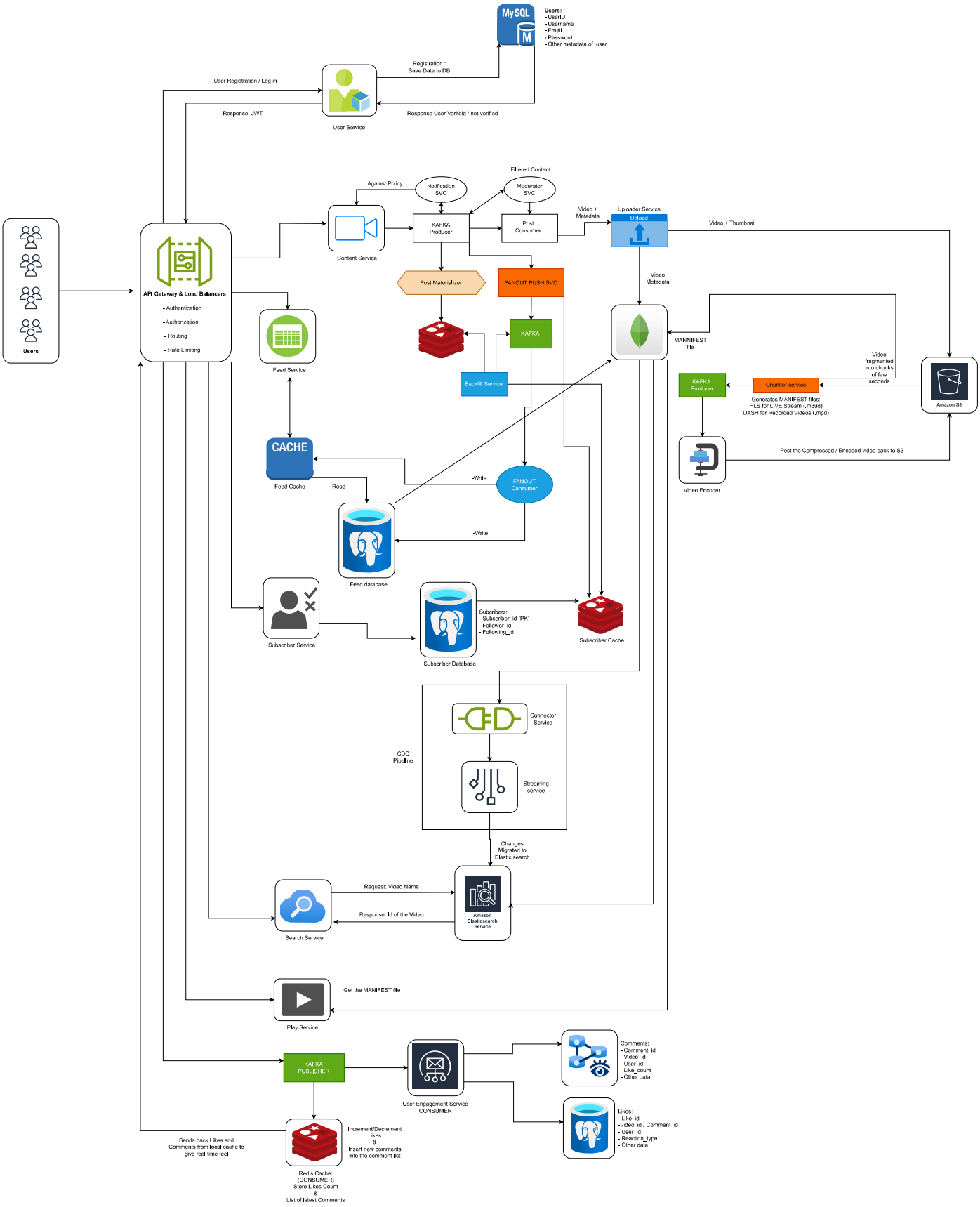
MySQL
Users:
- UserID
- Username
- Email
- Password
- Other metadata of user
Registration :
Save Data to DB
User Registration / Log in
Response: JWT
Response User Verifeid / not verified
User Service
Users
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Against Policy
Notification SVC
Filtered Content
Moderator SVC
Content Service
KAFKA Producer
Post Consumer
Video + Metadata
Uploader Service
Upload
Video + Thumbnail
Post Materializer
FANOUT PUSH SVC
Video Metadata
KAFKA
MANNIFEST file
Feed Service
Backfill Service
Video fragmented into chunks of few seconds
KAFKA Producer
Chunker service
Amazon S3
Generates MANIFEST files:
HLS for LIVE Stream (.m3u8)
DASH for Recorded Videos (.mpd)
CACHE
Feed Cache
-Read
-Write
FANOUT Consumer
Post the Compressed / Encoded video back to S3
Video Encoder
-Write
Feed database
Subscriber Service
Subscribers:
- Subscriber_id (PK)
- Follower_id
- Following_id
Subscriber Database
Subscriber Cache
Connector Service
CDC Pipeline
Streaming service
Changes Migrated to Elastic search
Request: Video Name
Response: Id of the Video
Search Service
Amazon Elasticsearch Service
Get the MANIFEST file
Play Service
KAFKA PUBLISHER
User Engagement Service CONSUMER
Comments:
- Comment_id
- Video_id
- User_id
- Like_count
- Other data
Likes:
- Like_id
-Video_id / Comment_id
- User_id
- Reaction_type
- Other data
Sends back Likes and Comments from local cache to give real time feel
Increment/Decrement Likes
&
Insert new comments into the comment list
Redis Cache:
(CONSUMER)
Store Likes Count
&
List of latest Comments